Robotic Car Painting Cell Simulation

TECHNICAL REPORT

Project: Robotics Simulation Intern – Practical Assessment

Name – Surname: Hend Khaled Haroun ABDELWAHAB

Department: Mechatronics Engineering

University: Yildiz Technical University

Location: Istanbul/Turkey

11 November 2026

YUKTII AI LABS

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1.1 Software Used

RoboDK (Free Trial Version) was used to develop the Robotic Car Painting Cell Simulation. RoboDK is a robot simulation and offline programming software known for its user-friendly interface and ease of use. It provides an extensive robot library, allowing users to select from a wide range of industrial robots. Additionally, the software supports various add-ins that simplify the design process and provide additional functionalities for robot simulation and programming.

2.1 Robot Selected

The **ABB IRB 5500-27** robot was selected for the car painting simulation (Figure 1). The IRB 5500-27 is a six-axis industrial robotic arm with a payload capacity of 16 kg and a maximum reach of 2,200 mm. It is specifically designed for automotive painting applications, including both interior and exterior surfaces, as well as other industrial painting operations.

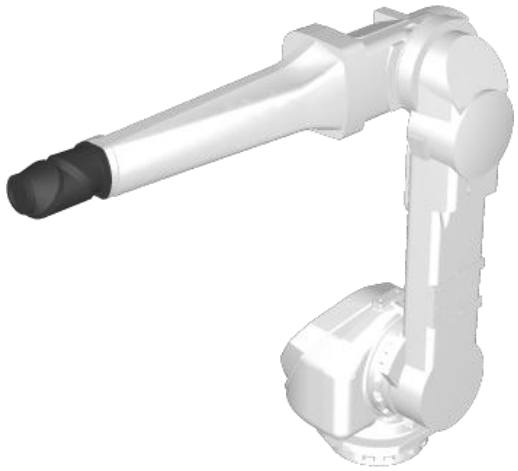


Figure1. ABB IRB 5500-27 Industrial Robot

3.1 Cell Layout

The cell layout was designed primarily to accommodate the main robotic simulation, which is located in a main room measuring 12 × 12 × 4 m (Figure 2).

The main cell consists of four primary components: the ABB robot, the car model, a turntable that serves as the car platform, and a pedestal that supports the robot.

The dimensions of the car model are as follows:

- **Length:** 4,826.62 mm
- **Width:** 1,894.88 mm
- **Height:** 1,599.26 mm

The turntable dimensions are:

- **Length:** 4,000 mm
- **Width:** 4,000 mm
- **Height:** 1,000 mm

The robot pedestal dimensions are:

- **Height:** 884 mm
- **Radius:** 250 mm

In addition to the main simulation room, the overall layout includes three additional rooms, each measuring 4 × 4 m. These rooms are reserved for other potential requirements and additional components of the robotic cell.

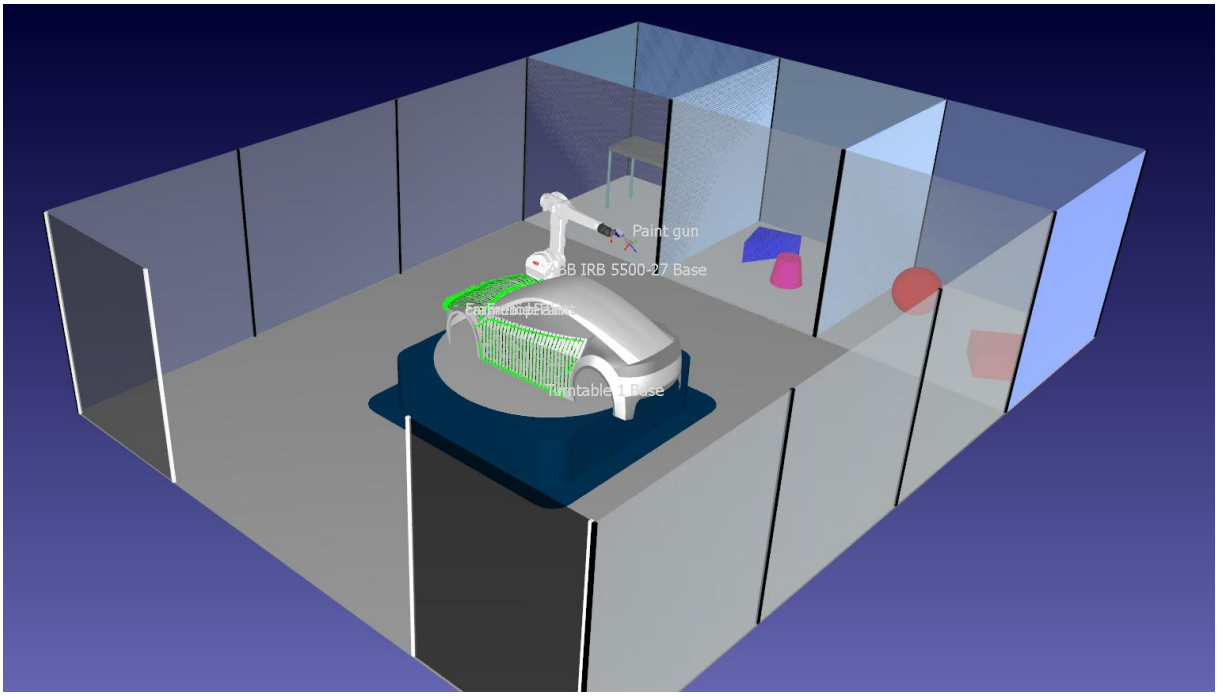


Figure 2. Robotic Car Painting Cell Layout.

4.1 Design Approach

For the robotic car painting cell simulation, a **Tesla Roadster CAD model** was selected as the target vehicle (Figure 3). The original car model was modified using SolidWorks to achieve the required dimensions.

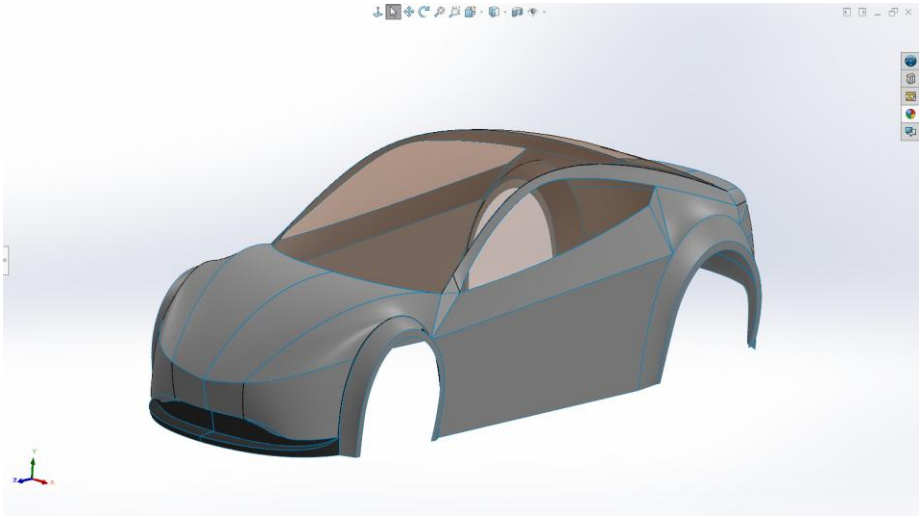


Figure 3. Tesla Roadster CAD Model

Both the selected ABB robot and the modified car model were then imported into RoboDK.

Since the main objective of the simulation is to paint the exterior body of the car, two possible approaches were considered:

1. Keeping the car stationary while the robot moves around it.
2. Keeping the robot stationary while the car rotates around its central axis.

The second approach was selected, in which the car rotates around its center while the robot remains stationary. This configuration was considered more suitable for the simulation because it simplifies the cell design and trajectory planning process.

A turntable was added to support and rotate the car, while a pedestal was designed to elevate and support the robot. Both components were created using the Components Add-in in RoboDK, and their dimensions were adjusted according to the car and robot models (Figures 4a and 4b).

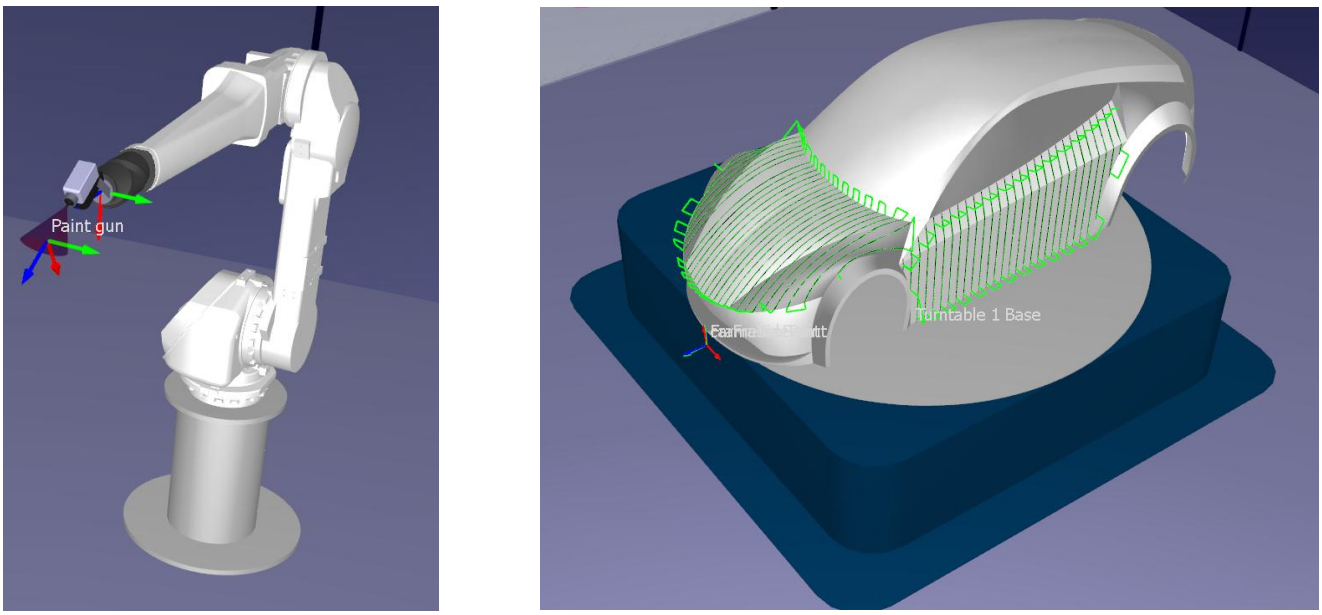


Figure 4 a) Robot Model and Pedestal (Robot Stand). b) Car Model and Turntable (Car Stand).

To generate the trajectory planning projects for the car painting application, curves were created in SolidWorks on two main target surfaces: the top and the sides of the car. These curves represent the desired painting paths that the robot must follow during the simulation.

5.1 Trajectory/Path Planning

The trajectory planning was developed in RoboDK using the **Curve Follow Project** feature. Separate path-planning projects were created for the two target surfaces: the top and the sides of the car.

The top painting operation was divided into two main sections. The first section covers the right half of the car's top surface, while the second covers the left half (Figures 5a and 5b). This division was implemented to organize the painting process and ensure that the entire target surface could be covered.

For the curve-following algorithm, **Minimum Tool Orientation Change** was selected to minimize unnecessary changes in the robot's tool orientation while following the generated curves.

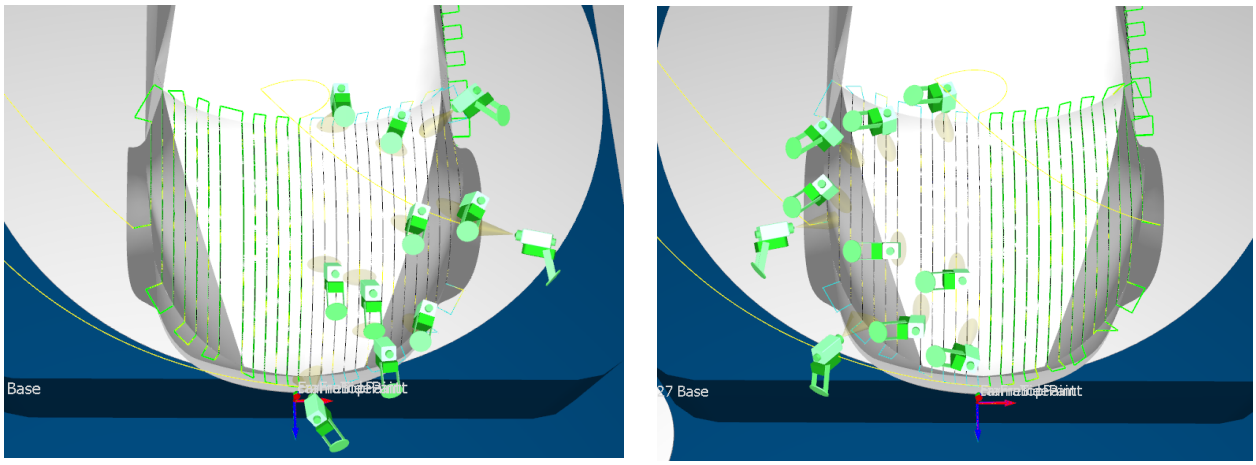


Figure 5a) Right-Side Car Painting Path Generated Using the Curve Follow Project. 5b) Left-Side Car Painting Path Generated Using the Curve Follow Project.

The main program, named **Top Paint**, combines the two painting sections into a single program (Figure 6). A connection between the two sections was established to ensure a continuous sequence of operations (Figure 7).

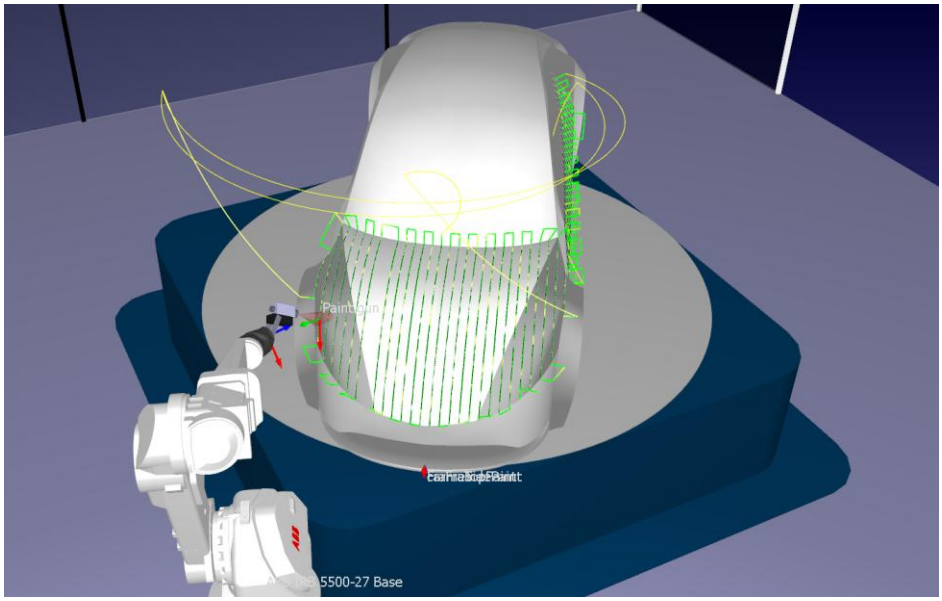


Figure 6. Top Paint Main Program – Curve Review.

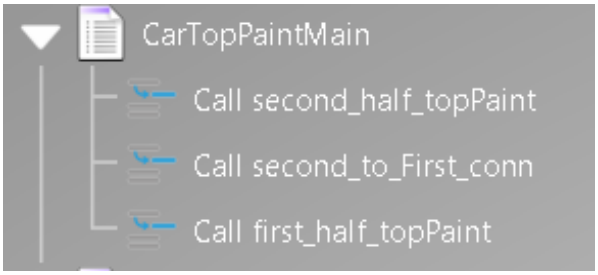


Figure 7. Top Paint Main Program – Instruction Sequence.

The side painting operation was developed using the same methodology. Curves were generated for the side surfaces of the car in SolidWorks and imported into RoboDK. The Curve Follow Project feature was then used to generate the corresponding robot paths.

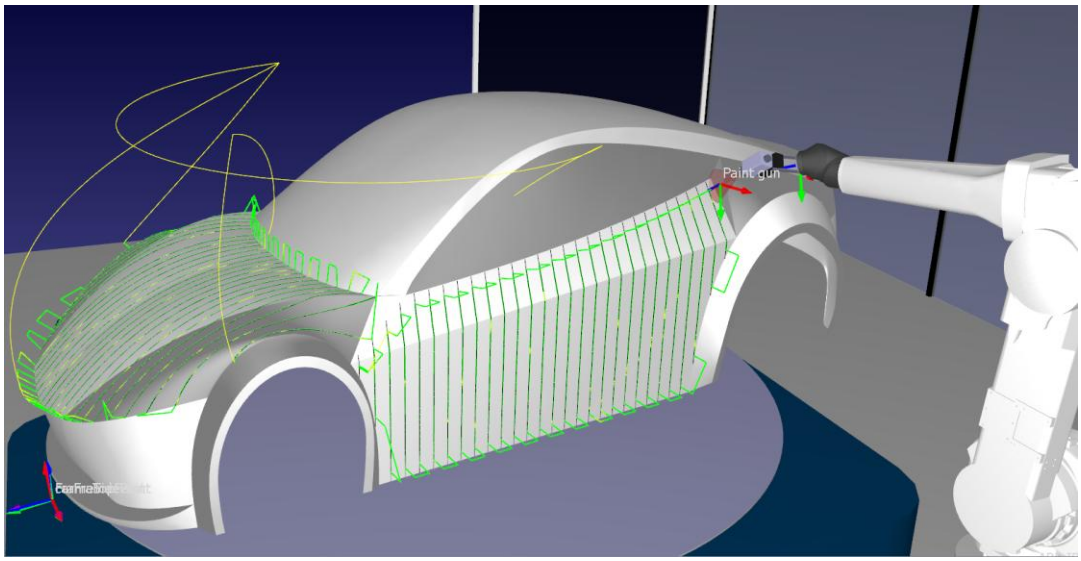


Figure 8. Side Paint Main Program – Curve Review.

As a result, two main painting programs were developed: Top Paint and Side Paint, covering the selected exterior surfaces of the vehicle.

6.1 Collision Checking

Collision checking was performed in RoboDK to identify potential collisions between the main components of the robotic cell, particularly between the robot and the car model.

This process was essential to ensure that the generated robot trajectories could be executed without unwanted physical interference between the components.

After the necessary adjustments were made to the cell layout and robot paths, no collisions were detected between the main components during the simulation (Figure 9).

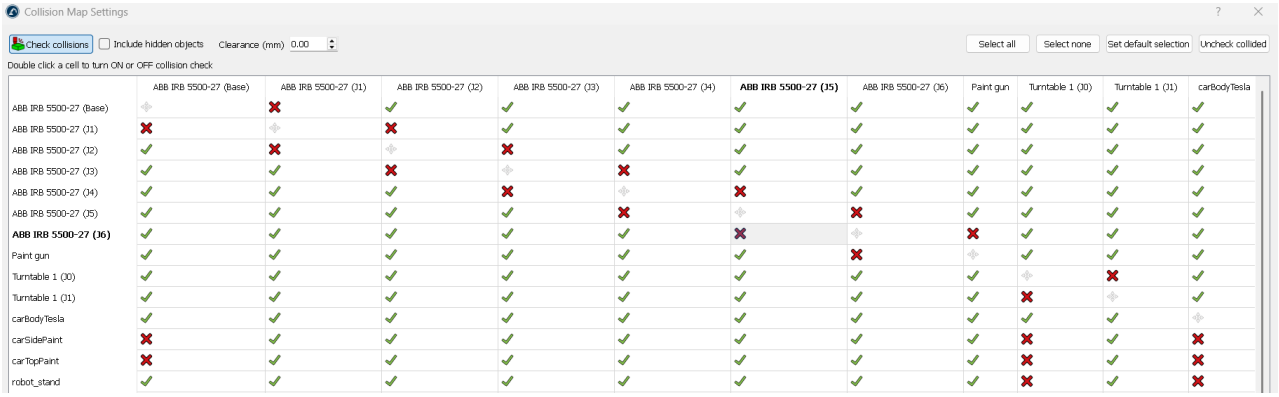


Figure 9. Collision Map of the Robotic Cell

7.1 Results

The Robotic Car Painting Cell Simulation was successfully developed using RoboDK. The final simulation includes the designed cell layout, modified car model, selected ABB robot, robot pedestal, and turntable.

Two main painting programs, **Top Paint** and **Side Paint**, were successfully generated. Both programs demonstrate the robot's ability to follow the predefined painting curves while the turntable rotates the car to provide access to the required painting surfaces.

Overall, the simulation demonstrates a functional robotic car painting cell in which the combination of robot trajectory planning and turntable rotation allows the selected exterior surfaces of the vehicle to be covered through an organized and reliable process.